

GROUP 1

HOSPITAL MANAGEMENT SYSTEM

Submitted by:

JHON NELLY D. MERCADO

KYLE R. CABRERA

CHRISTER LEM M. SIADOR

GODWIN TABONGBONG

MHARX NEVIN MALBOG

MA. LUIZITA MELENDEZ

Submitted to:

LANCE JERICO R. SALCEDO

Faculty, School of Information Technology Education
Universidad de Dagupan

HOSPITAL MANAGEMENT SYSTEM

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SECTION 1 – SYSTEM NARRATIVE

1.1 What the System Does

A Hospital Management System is a database-driven information system designed to manage and organize the core operational processes of a healthcare institution. The system centralizes patient, medical, administrative, and financial records into a single relational database to improve efficiency, consistency, and accessibility of hospital information.

The system supports essential hospital operations such as patient registration, appointment scheduling, inpatient admissions, room assignments, medical record management, prescription processing, billing, and discharge documentation. It also manages hospital personnel, including doctors and nurses, along with their departmental assignments and professional collaborations.

Using relational database principles, the system enforces data integrity through primary keys, foreign keys, unique constraints, and referential actions such as ON DELETE CASCADE, RESTRICT, and SET NULL. These mechanisms help reduce data redundancy, maintain consistency between related records, and ensure that hospital information remains accurate and reliable.

Additionally, the system allows healthcare personnel to efficiently retrieve patient histories, monitor room occupancy, track clinical activities, and maintain billing records for both inpatient and outpatient services. Overall, the Hospital Management System improves hospital workflow, resource management, and the organization of healthcare information.

1.2 Users and Stakeholders

- **Hospital Administrators** — Hospital administrators oversee the operational management of the hospital. They manage departmental information, supervise staff assignments, monitor room availability, and review billing and admission records to support administrative decision-making.
- **Doctors** — Doctors are responsible for diagnosing patients, creating medical records, prescribing medications, managing admissions, and preparing discharge reports. They also handle outpatient consultations through appointment scheduling.
- **Nurses** — Nurses assist doctors in patient care, clinical monitoring, and treatment support. They are assigned to departments and may collaborate with multiple doctors through the Doctor_Nurse relationship.

- **Patients** — Patients are the primary entities within the system. Their demographic, medical, admission, appointment, allergy, prescription, and billing information are stored and managed throughout the healthcare process
- **Billing Staff** — Billing personnel manage patient billing records, monitor payment statuses, and maintain financial documentation related to hospital services and admissions.
- **Records Officers** — Records officers maintain and organize patient documentation, including medical records, prescriptions, and discharge reports, ensuring data accuracy, consistency, and accessibility.

1.3 Why These Entities and Relationships Were Chosen

The entities and relationships in the system were designed to accurately represent real-world hospital operations while following relational database normalization principles.

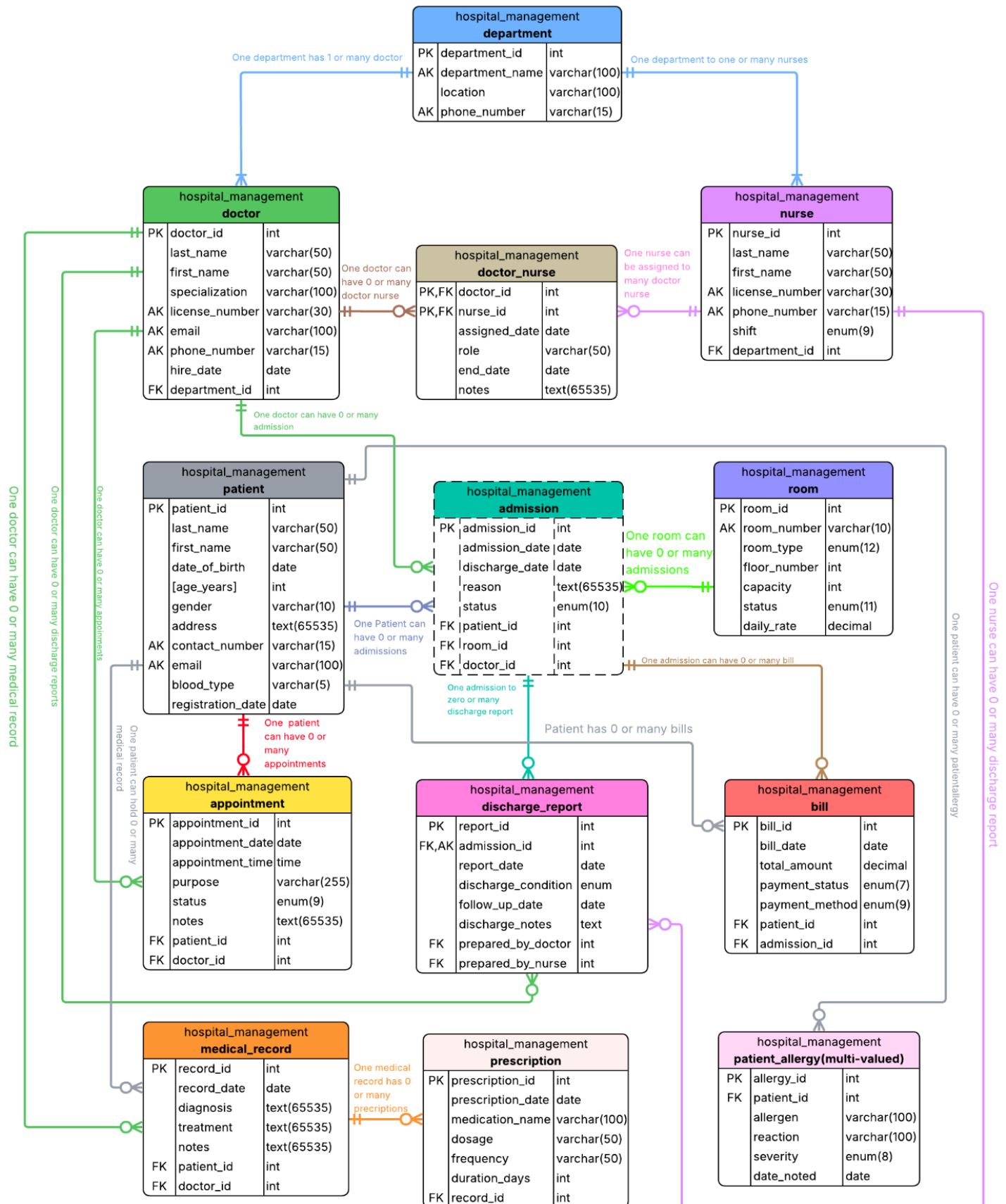
- **Department** — The Department entity serves as the organizational foundation of the hospital. Doctors and nurses are assigned to departments to represent hospital specialization and staff grouping.
- **Doctor & Nurse** — Doctors and nurses were modeled as separate entities because they have different responsibilities, credentials, and attributes. Their many-to-many relationship was resolved using the Doctor_Nurse bridge table, allowing multiple nurses to assist multiple doctors in clinical operations.
- **Patient** — The Patient entity acts as the core entity of the system since most operational records are directly associated with patients. Admissions, appointments, medical records, prescriptions, allergies, and billing records all depend on patient information.
- **PatientAllergy** — Patient allergies were separated into an independent entity to properly represent a multivalued attribute. A patient may have multiple allergies, and storing them in a separate table prevents redundancy and supports normalization.
- **Room** — Rooms were modeled independently to manage hospital resources such as room type, occupancy status, capacity, and daily rates. This allows efficient room allocation during patient admissions.
- **Admission (Weak Entity)** — The Admission entity represents inpatient hospital stays. It was treated conceptually as a weak entity because an admission cannot exist without an associated patient. Admissions also connect patients, doctors, and rooms during hospitalization.
- **Appointment** — Appointments represent scheduled outpatient consultations between patients and doctors. This entity was separated from Admission because outpatient consultations and inpatient admissions follow different operational processes.
- **MedicalRecord & Prescription** — MedicalRecord stores diagnoses, treatments, and clinical observations created by doctors for patients. Prescription was modeled as a dependent entity because one medical record may contain multiple prescribed medications.

- **Bill** — The Bill entity manages financial transactions related to patient services and hospital admissions. It supports both inpatient and outpatient billing through an optional relationship with Admission.
- **DischargeReport (1:1 with Admission)** — DischargeReport documents the final clinical summary of a completed admission. A one-to-one relationship was enforced using a UNIQUE constraint on admission_id to ensure that each admission can only have one discharge report.

1.4 Key Design Decisions

- **Derived Attribute (age_years)** — The age_years attribute was implemented as a virtual derived column using `TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE())`. This allows the patient's age to be automatically calculated from the date of birth instead of being manually stored, reducing redundancy and ensuring that the value always remains updated.
- **Multivalued Attribute (PatientAllergy)** — Allergies were implemented through a separate relational table rather than a single attribute within the Patient entity because a patient may have multiple allergic conditions. This design follows normalization principles and properly models the one-to-many relationship between Patient and PatientAllergy.
- **ON DELETE CASCADE** — Applied to relationships where child records are fully dependent on their parent entities, such as PatientAllergy, Admission, MedicalRecord, and Prescription. This ensures referential integrity by automatically deleting related child records when the parent record is removed.
- **ON DELETE RESTRICT** — Enforced on critical entities such as Doctor and Department to prevent deletion when related records still exist. This protects important hospital, clinical, and organizational data from accidental deletion.
- **ON DELETE SET NULL** — Used in the Bill entity for the admission_id foreign key to preserve billing records even if the associated admission record is deleted. This maintains historical financial information while removing the broken relationship reference.
- **ENUM Fields** — ENUM data types were implemented for controlled-value attributes such as room status, appointment status, payment status, discharge condition, room type, and nurse shift. This restricts invalid inputs and helps maintain consistent data throughout the system.
- **Bridge Table for Many-toMany Relationships** — The Doctor_Nurse table was created to resolve the many-to-many relationship between doctors and nurses. A composite primary key was used to ensure that each doctor-nurse assignment remains unique and properly normalized.

SECTION 2 – ERD DIAGRAM



SECTION 3 – RELATIONSHIP SUMMARY

The table presented below summarizes all defined relationships within the ERD, detailing their cardinality, participation constraints, and the corresponding SQL mechanisms implemented to enforce referential integrity and structural consistency.

Entities	Cardinality	Participation	SQL Enforcement
Doctor → Department	N:1	Total (Doctor)	FK department_id in Doctor
Nurse → Department	N:1	Total (Nurse)	FK department_id in Nurse
Patient → PatientAllergy	1:N	Partial (Patient)	FK patient_id in PatientAllergy
Patient → Admission	1:N	Total (Admission)	FK patient_id in Admission + CASCADE
Admission → Room	N:1	Total (Admission)	FK room_id in Admission
Doctor → Admission	1:N	Partial (Doctor)	FK doctor_id in Admission
Patient ↔ Doctor (Appointment)	M:N	Partial (both)	FKs in Appointment table
Doctor → MedicalRecord	1:N	Partial (Doctor)	FK doctor_id in MedicalRecord
Patient → MedicalRecord	1:N	Partial (Patient)	FK patient_id in MedicalRecord
MedicalRecord → Prescription	1:N	Partial (MedicalRecord)	FK record_id in Prescription
Patient → Bill	1:N	Partial (Patient)	FK patient_id in Bill
Doctor ↔ Nurse	M:N	Partial (both)	Bridge table Doctor_Nurse
Admission ↔ DischargeReport	1:N	Partial (Admission)	FK + UNIQUE on admission_id

SECTION 4 – DATA DICTIONARY

4.1 Department

Stores hospital departments. No foreign keys — root entity.

Column	Data Type	Constraints	Description
department_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique department identifier
department_name	VARCHAR(100)	NOT NULL, UNIQUE	Full name of the department
location	VARCHAR(100)	NOT NULL	Physical building / floor location
phone_number	VARCHAR(15)	UNIQUE	Department contact number

4.2 Doctor

Medical staff. References Department (1 doctor → 1 department)

Column	Data Type	Constraints	Description
doctor_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique doctor identifier
last_name	VARCHAR(50)	NOT NULL	Doctor's surname
first_name	VARCHAR(50)	NOT NULL	Doctor's given name
specialization	VARCHAR(100)	NOT NULL	Medical field / specialty
license_number	VARCHAR(30)	NOT NULL, UNIQUE	PRC license number
email	VARCHAR(100)	UNIQUE	Professional email address
phone_number	VARCHAR(15)	UNIQUE	Mobile contact number
hire_date	DATE	NOT NULL	Date of employment
department_id	INT	FK → Department, NOT NULL	Department the doctor belongs to

4.3 Patient

Registered patients. Contains derived attribute age_years

Column	Data Type	Constraints	Description
patient_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique patient identifier
last_name	VARCHAR(50)	NOT NULL	Patient's surname
first_name	VARCHAR(50)	NOT NULL	Patient's given name
date_of_birth	DATE	NOT NULL	Birth date (used to derive age)
age_years	INT (VIRTUAL)	DERIVED — computed via TIMESTAMPDIFF	Current age in years (not stored)
gender	VARCHAR(10)	NOT NULL	Patient's gender
address	TEXT	NULLABLE	Home address
contact_number	VARCHAR(15)	UNIQUE	Mobile contact number
email	VARCHAR(100)	UNIQUE	Personal email address
blood_type	VARCHAR(5)	NULLABLE	ABO/Rh blood type
registration_date	DATE	NOT NULL, DEFAULT CURRENT_DATE	Date of hospital registration

4.4 PatientAllergy

MULTIVALUED attribute — a patient may have multiple allergies

Column	Data Type	Constraints	Description
allergy_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique allergy record identifier
patient_id	INT	FK → Patient, NOT NULL	Owning patient
allergen	VARCHAR(100)	NOT NULL	Substance causing the allergy

reaction	VARCHAR(100)	NOT NULL	Observed allergic reaction
severity	ENUM	NOT NULL, DEFAULT 'Mild'	Mild / Moderate / Severe
date_noted	DATE	NOT NULL	Date the allergy was recorded

4.5 Room

Hospital rooms available for patient admission.

Column	Data Type	Constraints	Description
room_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique room identifier
room_number	VARCHAR(10)	NOT NULL, UNIQUE	Room code / number
room_type	ENUM	NOT NULL	ICU / General / Private / Semi-Private / Public
floor_number	INT	NOT NULL	Floor the room is located on
capacity	INT	NOT NULL, DEFAULT 1	Maximum number of patients
status	ENUM	NOT NULL, DEFAULT 'Available'	Available / Occupied / Maintenance
daily_rate	DECIMAL(10,2)	NOT NULL	Charge per day in Philippine Peso

4.6 Nurse

Nursing staff. References Department

Column	Data Type	Constraints	Description
nurse_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique nurse identifier
last_name	VARCHAR(50)	NOT NULL	Nurse's surname
first_name	VARCHAR(50)	NOT NULL	Nurse's given name

license_number	VARCHAR(30)	NOT NULL, UNIQUE	PRC license number
phone_number	VARCHAR(15)	UNIQUE	Mobile contact number
shift	ENUM	NOT NULL	Morning / Afternoon / Night
department_id	INT	FK → Department, NOT NULL	Department the nurse is assigned to

4.7 Admission

WEAK ENTITY — existentially dependent on Patient. Records inpatient stays

Column	Data Type	Constraints	Description
admission_id	INT	PK, NOT NULL, AUTO_INCREMENT	Surrogate PK (implementation convenience)
admission_date	DATE	NOT NULL	Date patient was admitted
discharge_date	DATE	NULLABLE	Date patient was discharged (NULL if still admitted)
reason	TEXT	NOT NULL	Clinical reason for admission
status	ENUM	NOT NULL, DEFAULT 'Admitted'	Admitted / Discharged
patient_id	INT	FK → Patient, NOT NULL	Patient being admitted
room_id	INT	FK → Room, NOT NULL	Room assigned to the patient
doctor_id	INT	FK → Doctor, NOT NULL	Attending physician

4.8 Appointment

Scheduled consultations between a patient and a doctor.

Column	Data Type	Constraints	Description
appointment_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique appointment identifier
appointment_date	DATE	NOT NULL	Scheduled date
appointment_time	TIME	NOT NULL	Scheduled date
purpose	VARCHAR(255)	NOT NULL	Reason for the appointment
status	ENUM	NOT NULL, DEFAULT 'Pending'	Pending / Completed / Cancelled
notes	TEXT	NULLABLE	Additional remarks
patient_id	INT	FK → Patient, NOT NULL	Patient attending
doctor_id	INT	FK → Doctor, NOT NULL	Doctor being consulted

4.9 MedicalRecord

Clinical records created by a doctor for a patient visit.

Column	Data Type	Constraints	Description
record_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique record identifier
record_date	DATE	NOT NULL	Date the record was created
diagnosis	TEXT	NOT NULL	Doctor's clinical diagnosis
treatment	TEXT	NULLABLE	Prescribed treatment plan
notes	TEXT	NULLABLE	Additional clinical notes
patient_id	INT	FK → Patient, NOT NULL	Patient this record belongs to

doctor_id	INT	FK → Doctor, NOT NULL	Doctor who created the record
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4.10 Prescription

Medication orders linked to a specific medical record.

Column	Data Type	Constraints	Description
prescription_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique prescription identifier
prescription_date	DATE	NOT NULL	Date the prescription was written
medication_name	VARCHAR(100)	NOT NULL	Name of the prescribed medicine
dosage	VARCHAR(50)	NOT NULL	Strength / amount per dose
frequency	VARCHAR(50)	NOT NULL	How often it is taken
duration_days	INT	NOT NULL	Total number of days to take medication
record_id	INT	FK → MedicalRecord, NOT NULL	Medical record this prescription belongs to

4.11 Bill

Financial billing record for a patient's admission or visit

Column	Data Type	Constraints	Description
bill_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique billing identifier
bill_date	DATE	NOT NULL	Date the bill was generated

total_amount	DECIMAL(12,2)	NOT NULL	Total amount charged in Philippine Peso
payment_status	ENUM	NOT NULL, DEFAULT 'Unpaid	Unpaid / Paid / Partial
payment_method	ENUM	NULLABLE	Cash / Card / Insurance
patient_id	INT	FK → Patient, NOT NULL	Patient being billed
admission_id	INT	FK → Admission, NULLABLE	Associated admission (NULL for outpatient)

4.12 Doctor_Nurse

BRIDGE TABLE (M:N) — resolves many-to-many between Doctor and Nurse.

Column	Data Type	Constraints	Description
doctor_id	INT	PK (composite), FK → Doctor	Doctor in the assignment
nurse_id	INT	PK (composite), FK → Nurse	Nurse in the assignment
assigned_date	DATE	NOT NULL	Date the assignment started
role	VARCHAR(50)	NOT NULL, DEFAULT 'Clinical Support'	Role of the nurse under this doctor
end_date	DATE	NULLABLE	Date the assignment ended (NULL if ongoing)
notes	TEXT	NULLABLE	Additional assignment notes

4.13 DischargeReport

1:1 with Admission — one report is produced per completed admission

Column	Data Type	Constraints	Description
report_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique discharge report identifier
admission_id	INT	FK → Admission, NOT NULL, UNIQUE	Linked admission (UNIQUE enforces 1:1)
report_date	DATE	NOT NULL	Date the report was prepared
discharge_condition	ENUM	NOT NULL	Stable / Critical / Deceased / Transferred
follow_up_date	DATE	NULLABLE	Recommended follow-up consultation date
discharge_notes	TEXT	NULLABLE	Summary notes at time of discharge
prepared_by_doctor	INT	FK → Doctor, NOT NULL	Doctor who signed the discharge report
prepared_by_nurse	INT	FK → Nurse, NULLABLE	Nurse who assisted in preparing the report

SECTION 5 – SCREENSHOT OF QUERIES

```
1  -- DATABASE CREATION
2  CREATE DATABASE IF NOT EXISTS hospital_management;
3  USE hospital_management;

5  -- TABLE: Department
6  -- 1 PK, 0 FK
7  CREATE TABLE Department (
8      department_id INT NOT NULL AUTO_INCREMENT,
9      department_name VARCHAR(100) NOT NULL UNIQUE,
10     location VARCHAR(100) NOT NULL,
11     phone_number VARCHAR(15) UNIQUE,
12 );

14 -- TABLE: Doctor
15 -- 1 PK, 1 FK
16 CREATE TABLE Doctor (
17     doctor_id INT NOT NULL AUTO_INCREMENT,
18     last_name VARCHAR(50) NOT NULL,
19     first_name VARCHAR(50) NOT NULL,
20     specialization VARCHAR(100) NOT NULL,
21     license_number VARCHAR(30) NOT NULL UNIQUE,
22     email VARCHAR(100) UNIQUE,
23     phone_number VARCHAR(15) UNIQUE,
24     hire_date DATE NOT NULL,
25     department_id INT NOT NULL,
26     PRIMARY KEY (doctor_id),
27     FOREIGN KEY (department_id) REFERENCES Department (department_id)
28     ON DELETE RESTRICT ON UPDATE CASCADE
29 );
```

```
32  -- TABLE: Patient
33  -- 1 PK, 0 FK
34  -- age_years is DERIVED and was computed from the date_of_birth
35  CREATE TABLE Patient (
36      patient_id INT NOT NULL AUTO_INCREMENT,
37      last_name VARCHAR(50) NOT NULL,
38      first_name VARCHAR(50) NOT NULL,
39      date_of_birth DATE NOT NULL,
40      age_years INT AS (TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE())) VIRTUAL,
41      gender VARCHAR(10) NOT NULL,
42      address TEXT,
43      contact_number VARCHAR(15) UNIQUE,
44      email VARCHAR(100) UNIQUE,
45      blood_type VARCHAR(5),
46      registration_date DATE NOT NULL DEFAULT (CURRENT_DATE),
47      PRIMARY KEY (patient_id)
48  );

50  -- TABLE: Patient Allergy (MULTIVALUED)
51  -- 1 PK, 0 FK
52  CREATE TABLE PatientAllergy (
53      allergy_id INT NOT NULL AUTO_INCREMENT,
54      patient_id INT NOT NULL,
55      allergen VARCHAR(100) NOT NULL,
56      reaction VARCHAR(100) NOT NULL,
57      severity ENUM('Mild', 'Moderate', 'Severe') NOT NULL DEFAULT 'Mild',
58      date_noted DATE NOT NULL,
59      PRIMARY KEY (allergy_id),
60      FOREIGN KEY (patient_id) REFERENCES Patient (patient_id)
61      | ON DELETE CASCADE ON UPDATE CASCADE
62  );

64  -- TABLE: ROOM
65  -- 1 PK, 0 FK
66  CREATE TABLE Room (
67      room_id INT NOT NULL AUTO_INCREMENT,
68      room_number VARCHAR(10) NOT NULL UNIQUE,
69      room_type ENUM('ICU', 'General', 'Private', 'Semi-Private', 'Public') NOT NULL,
70      floor_number INT NOT NULL,
71      capacity INT NOT NULL DEFAULT 1,
72      status ENUM('Available', 'Occupied', 'Maintenance') NOT NULL DEFAULT 'Available',
73      daily_rate DECIMAL(10, 2) NOT NULL,
74      PRIMARY KEY (room_id)
75  );
```



```
77  -- TABLE: NURSE
78  -- 1 PK, 1 FK
79  CREATE TABLE Nurse (
80      nurse_id INT NOT NULL AUTO_INCREMENT,
81      last_name VARCHAR(50) NOT NULL,
82      first_name VARCHAR(50) NOT NULL,
83      license_number VARCHAR(30) NOT NULL UNIQUE,
84      phone_number VARCHAR(15) UNIQUE,
85      shift ENUM('Morning', 'Afternoon', 'Night') NOT NULL,
86      department_id INT NOT NULL,
87      PRIMARY KEY (nurse_id),
88      FOREIGN KEY (department_id) REFERENCES Department (department_id)
89      | ON DELETE RESTRICT ON UPDATE CASCADE
90  );

92  -- TABLE: Admission (WEAK ENTITY)
93  -- 1 PK, 3 FK
94  CREATE TABLE Admission (
95      admission_id INT NOT NULL AUTO_INCREMENT,
96      admission_date DATE NOT NULL,
97      discharge_date DATE,
98      reason TEXT NOT NULL,
99      status ENUM('Admitted', 'Discharged') NOT NULL DEFAULT 'Admitted',
100     patient_id INT NOT NULL,
101     room_id INT NOT NULL,
102     doctor_id INT NOT NULL,
103     PRIMARY KEY (admission_id),
104     FOREIGN KEY (patient_id) REFERENCES Patient (patient_id)
105     | ON DELETE CASCADE ON UPDATE CASCADE,
106     FOREIGN KEY (room_id) REFERENCES Room (room_id)
107     | ON DELETE RESTRICT ON UPDATE CASCADE,
108     FOREIGN KEY (doctor_id) REFERENCES Doctor (doctor_id)
109     | ON DELETE RESTRICT ON UPDATE CASCADE
110 );
```

```
112  -- TABLE: Appointment
113  -- 1 PK, 2 FK
114  CREATE TABLE Appointment (
115      appointment_id INT NOT NULL AUTO_INCREMENT,
116      appointment_date DATE NOT NULL,
117      appointment_time TIME NOT NULL,
118      purpose VARCHAR(255) NOT NULL,
119      status ENUM('Pending', 'Completed', 'Cancelled') NOT NULL DEFAULT 'Pending',
120      notes TEXT,
121      patient_id INT NOT NULL,
122      doctor_id INT NOT NULL,
123      PRIMARY KEY (appointment_id),
124      FOREIGN KEY (patient_id) REFERENCES Patient (patient_id)
125      | ON DELETE CASCADE ON UPDATE CASCADE,
126      FOREIGN KEY (doctor_id) REFERENCES Doctor (doctor_id)
127      | ON DELETE RESTRICT ON UPDATE CASCADE
128  );
```

```
130  -- TABLE: MedicalRecord
131  -- 1 PK, 2 FK
132  CREATE TABLE MedicalRecord (
133      record_id INT NOT NULL AUTO_INCREMENT,
134      record_date DATE NOT NULL,
135      diagnosis TEXT NOT NULL,
136      treatment TEXT,
137      notes TEXT,
138      patient_id INT NOT NULL,
139      doctor_id INT NOT NULL,
140      PRIMARY KEY (record_id),
141      FOREIGN KEY (patient_id) REFERENCES Patient (patient_id)
142      | ON DELETE CASCADE ON UPDATE CASCADE,
143      FOREIGN KEY (doctor_id) REFERENCES Doctor (doctor_id)
144      | ON DELETE RESTRICT ON UPDATE CASCADE
145  );
```

```
147  -- TABLE: Prescription
148  -- 1 PK, 1 FK
149  CREATE TABLE Prescription (
150      prescription_id INT NOT NULL AUTO_INCREMENT,
151      prescription_date DATE NOT NULL,
152      medication_name VARCHAR(100) NOT NULL,
153      dosage VARCHAR(50) NOT NULL,
154      frequency VARCHAR(50) NOT NULL,
155      duration_days INT NOT NULL,
156      record_id INT NOT NULL,
157      PRIMARY KEY (prescription_id),
158      FOREIGN KEY (record_id) REFERENCES MedicalRecord (record_id)
159      | ON DELETE CASCADE ON UPDATE CASCADE
160  );

162  -- TABLE: Bill
163  -- 1 PK, 2 FK
164  CREATE TABLE Bill (
165      bill_id INT NOT NULL AUTO_INCREMENT,
166      bill_date DATE NOT NULL,
167      total_amount DECIMAL(12, 2) NOT NULL,
168      payment_status ENUM('Unpaid', 'Paid', 'Partial') NOT NULL DEFAULT 'Unpaid',
169      payment_method ENUM('Cash', 'Card', 'Insurance'),
170      patient_id INT NOT NULL,
171      admission_id INT,
172      PRIMARY KEY (bill_id),
173      FOREIGN KEY (patient_id) REFERENCES Patient (patient_id)
174      | ON DELETE RESTRICT ON UPDATE CASCADE,
175      FOREIGN KEY (admission_id) REFERENCES Admission (admission_id)
176      | ON DELETE SET NULL ON UPDATE CASCADE
177  );
```

```
179  -- TABLE: Doctor_Nurse (BRIDGE TABLE)
180  -- 1 PK, 2 FK
181  -- HAVE COMPOSITE PRIMARY KEY
182  CREATE TABLE Doctor_Nurse (
183      doctor_id INT NOT NULL,
184      nurse_id INT NOT NULL,
185      assigned_date DATE NOT NULL,
186      role VARCHAR(50) NOT NULL DEFAULT 'Clinical Support',
187      end_date DATE,
188      notes TEXT,
189      PRIMARY KEY (doctor_id, nurse_id),
190      FOREIGN KEY (doctor_id) REFERENCES Doctor (doctor_id)
191      | ON DELETE CASCADE ON UPDATE CASCADE,
192      FOREIGN KEY (nurse_id) REFERENCES Nurse (nurse_id)
193      | ON DELETE CASCADE ON UPDATE CASCADE
194  );

196  -- TABLE: DischargeReport
197  -- 1 PK, 1 FK
198  CREATE TABLE DischargeReport (
199      report_id INT NOT NULL AUTO_INCREMENT,
200      admission_id INT NOT NULL UNIQUE,
201      report_date DATE NOT NULL,
202      discharge_condition ENUM('Stable', 'Critical', 'Deceased', 'Transferred') NOT NULL,
203      follow_up_date DATE,
204      discharge_notes TEXT,
205      prepared_by_doctor INT NOT NULL,
206      prepared_by_nurse INT NULL,
207      PRIMARY KEY (report_id),
208      FOREIGN KEY (admission_id) REFERENCES Admission (admission_id)
209      | ON DELETE CASCADE ON UPDATE CASCADE,
210      FOREIGN KEY (prepared_by_doctor) REFERENCES Doctor (doctor_id)
211      | ON DELETE RESTRICT ON UPDATE CASCADE,
212      FOREIGN KEY (prepared_by_nurse) REFERENCES Nurse (nurse_id)
213      | ON DELETE RESTRICT ON UPDATE CASCADE
214  );
```

```
218  -- INSERT: Department
219  INSERT INTO Department
220  (department_name, location, phone_number)
221  VALUES
222  ('Cardiology', 'Building A, Floor 2', '09670679301'),
223  ('Neurology', 'Building B, Floor 3', '09670679302'),
224  ('Orthopedics', 'Building A, Floor 1', '09670679303'),
225  ('Pediatrics', 'Building C, Floor 1', '09670679304'),
226  ('Emergency Medicine', 'Building D, Floor 1', '09670679305');

228  -- INSERT: Doctor
229  INSERT INTO Doctor
230  (last_name, first_name, specialization, license_number, email, phone_number,
231  hire_date, department_id)
232  VALUES
233  ('Tsukishima', 'Kei', 'Cardiologist', 'PRC-MD-67021', 'tsukki@gmail.com',
234  '0923710501', '2018-09-27', 1),
235  ('Tomioka', 'Giyu', 'Neurologist', 'PRC-MD-67022', 'giyu@gmail.com',
236  '0923710502', '2019-02-08', 2),
237  ('Rukawa', 'Kaede', 'Orthopedic Surgeon', 'PRC-MD-67023', 'rukawa@gmail.com',
238  '0923710503', '2020-01-01', 3),
239  ('Nanami', 'Kento', 'Pediatrician', 'PRC-MD-67024', 'nanami@gmail.com',
240  '0923710504', '2021-07-03', 4),
241  ('Vinsmoke', 'Sanji', 'Emergency Physician', 'PRC-MD-67025', 'sanji@gmail.com',
242  '0923710505', '2022-03-12', 5);

238  -- INSERT: Patient
239  INSERT INTO Patient
240  (last_name, first_name, date_of_birth, gender, address, contact_number, email,
241  blood_type, registration_date)
242  VALUES
243  ('Freeccs', 'Gon', '2001-05-05', 'Male', 'Whale Island, Yorknew City',
244  '09766706721', 'gon@gmail.com', 'O+', '2026-01-10'),
245  ('Zoldyck', 'Killua', '2001-07-07', 'Male', 'Kukuroo Mountain, Padokea',
246  '09766706722', 'killua@gmail.com', 'A+', '2026-01-11'),
247  ('Lucilfer', 'Chrollo', '1988-02-15', 'Male', 'Meteor City', '09766706723',
248  'chrollo@gmail.com', 'B+', '2026-01-12'),
249  ('Morrow', 'Hisoka', '1990-06-06', 'Male', 'Heavens Arena, Republic of Padokea',
250  '09766706724', 'hisoka@gmail.com', 'O-', '2026-01-13'),
251  ('Zoldyck', 'Alluka', '2003-09-09', 'Female', 'Kukuroo Mountain, Padokea',
252  '09766706725', 'alluka@gmail.com', 'B-', '2026-01-14');
```

```
248  -- INSERT: Room
249  INSERT INTO Room
250  (room_number, room_type, floor_number, capacity, status, daily_rate)
251  VALUES
252  ('101', 'General', 1, 4, 'Occupied', 500.00),
253  ('202', 'Private', 2, 1, 'Occupied', 2500.00),
254  ('303', 'ICU', 3, 2, 'Occupied', 5000.00),
255  ('104', 'Semi-Private', 1, 2, 'Available', 1200.00),
256  ('205', 'Public', 2, 6, 'Available', 300.00);

258  -- INSERT: Nurse
259  INSERT INTO Nurse
260  (last_name, first_name, license_number, phone_number, shift, department_id)
261  VALUES
262  ('Kim', 'Sa-bu', 'RN-67051', '09171234565', 'Morning', 1),
263  ('Seo', 'Woo-jin', 'RN-67052', '09171234566', 'Afternoon', 2),
264  ('Cha', 'Eun-jae', 'RN-67053', '09171234567', 'Night', 3),
265  ('Yoon', 'A-reum', 'RN-67054', '09171234568', 'Morning', 4),
266  ('Park', 'Min-guk', 'RN-67055', '09171234569', 'Afternoon', 5);

268  -- INSERT: PatientAllergy
269  INSERT INTO PatientAllergy
270  (patient_id, allergen, reaction, severity, date_noted)
271  VALUES
272  (1, 'Penicillin', 'Skin rash and hives', 'Moderate', '2026-01-06'),
273  (1, 'Peanuts', 'Difficulty breathing', 'Severe', '2026-01-07'),
274  (1, 'Dust', 'Sneezing and itchy eyes', 'Mild', '2026-01-08'),
275  (2, 'Aspirin', 'Gastric bleeding', 'Severe', '2026-02-15'),
276  (2, 'Seafood', 'Swollen lips', 'Moderate', '2026-02-18'),
277  (3, 'Milk', 'Sneezing and watery eyes', 'Mild', '2026-03-02'),
278  (4, 'Eggs', 'Skin irritation', 'Moderate', '2026-03-05'),
279  (5, 'Mold', 'Sneezing and coughing', 'Mild', '2026-04-13'),
280  (5, 'Perfume', 'Headache and skin irritation', 'Mild', '2026-04-15');
```

```
282  -- INSERT: Admission
283  INSERT INTO Admission
284  (admission_date, discharge_date, reason, status, patient_id, room_id, doctor_id)
285  VALUES
286  ('2026-04-01', '2026-04-07', 'Chest pain and shortness of breath', 'Discharged',
287   1, 2, 1),
288  ('2026-04-05', '2026-04-12', 'Severe migraine with vomiting', 'Discharged', 2, 1,
289   2),
290  ('2026-04-10', NULL, 'Post-surgical knee monitoring', 'Admitted', 3, 3, 3),
291  ('2026-04-15', '2026-04-20', 'High fever and febrile seizure', 'Discharged', 4,
292   4, 4),
293  ('2026-04-18', NULL, 'Traumatic head injury', 'Admitted', 5, 3, 5);

294  -- INSERT: Appointment
295  INSERT INTO Appointment
296  (appointment_date, appointment_time, purpose, status, notes, patient_id, doctor_id)
297  VALUES
298  ('2026-05-02', '09:00:00', 'Cardiac follow-up check', 'Completed', 'ECG and BP
299   monitoring done', 1, 1),
300  ('2026-05-05', '10:30:00', 'Neurology consultation', 'Completed', 'MRI results
301   reviewed', 2, 2),
302  ('2026-05-08', '14:00:00', 'Orthopedic rehabilitation eval', 'Pending', NULL, 3,
303   3),
304  ('2026-05-10', '08:00:00', 'Pediatric growth assessment', 'Cancelled', 'Patient
305   rescheduled', 4, 4),
306  ('2026-05-15', '11:00:00', 'Emergency medicine follow-up', 'Pending', NULL, 5, 5);

307  -- INSERT: Medical Record
308  INSERT INTO MedicalRecord
309  (record_date, diagnosis, treatment, notes, patient_id, doctor_id)
310  VALUES
311  ('2026-04-07', 'Stable angina', 'Nitrates and beta-blockers prescribed', 'Monitor
312   BP weekly', 1, 1),
313  ('2026-04-12', 'Chronic migraine', 'Topiramate prescribed, rest advised', 'Avoid
314   screen exposure', 2, 2),
315  ('2026-04-10', 'Post-op knee arthroscopy', 'Physical therapy and pain
316   management', 'Weight-bearing restricted', 3, 3),
317  ('2026-04-18', 'Febrile seizure', 'IV fluids and antipyretics administered',
318   'Follow-up EEG scheduled', 4, 4),
319  ('2026-04-20', 'Traumatic brain injury', 'CT scan ordered, ICU monitoring
320   initiated', 'GCS score 12 on admission', 5, 5);
```



```
312  -- INSERT: Prescription
313  INSERT INTO Prescription
314  (prescription_date, medication_name, dosage, frequency, duration_days, record_id)
315  VALUES
316  ('2026-04-07', 'Isosorbide Mononitrate', '20mg', 'Twice daily', 30, 1),
317  ('2026-04-12', 'Topiramate', '50mg', 'Once daily at night', 60, 2),
318  ('2026-04-10', 'Celecoxib', '200mg', 'Once daily', 14, 3),
319  ('2026-04-18', 'Paracetamol', '500mg', 'Every 6 hours', 5, 4),
320  ('2026-04-20', 'Mannitol', '100g', 'IV once daily', 7, 5);

322  -- INSERT: Bill
323  INSERT INTO Bill
324  (bill_date, total_amount, payment_status, payment_method, patient_id, admission_id)
325  VALUES
326  ('2026-04-07', 18500.00, 'Paid', 'Insurance', 1, 1),
327  ('2026-04-12', 12000.00, 'Paid', 'Cash', 2, 2),
328  ('2026-04-15', 35000.00, 'Partial', 'Card', 3, 3),
329  ('2026-04-19', 8750.00, 'Paid', 'Cash', 4, 4),
330  ('2026-04-29', 42000.00, 'Unpaid', NULL, 5, 5);

332  -- INSERT: Doctor_Nurse
333  INSERT INTO Doctor_Nurse
334  (doctor_id, nurse_id, assigned_date, role, end_date, notes)
335  VALUES
336  (1, 1, '2026-01-01', 'Clinical Support', NULL, 'Assigned to cardiuuology ward'),
337  (1, 2, '2026-01-02', 'Clinical Support', NULL, 'Assists during rounds'),
338  (2, 2, '2026-01-01', 'Clinical Support', NULL, 'Neurology assistance'),
339  (2, 3, '2026-01-01', 'Clinical Support', NULL, 'MRI and patient monitoring'),
340  (3, 3, '2026-01-01', 'Surgical Support', NULL, 'OR support team'),
341  (3, 1, '2026-01-01', 'Surgical Support', NULL, 'Post-op assistance'),
342  (4, 4, '2026-01-01', 'Clinical Support', NULL, 'Pediatric ward coverage'),
343  (4, 5, '2026-01-01', 'Clinical Support', NULL, 'Child patient monitoring'),
344  (5, 5, '2026-01-01', 'Triage Support', NULL, 'ER triage'),
345  (5, 2, '2026-01-01', 'Triage Support', NULL, 'Emergency assistance');

347  -- INSERT: DischargeReport
348  INSERT INTO DischargeReport
349  (admission_id, report_date, discharge_condition, follow_up_date, discharge_notes,
350  prepared_by_doctor, prepared_by_nurse)
351  VALUES
352  (1, '2026-04-07', 'Stable', '2026-05-02', 'Patient discharged in stable
353  condition. Continue medications as prescribed.', 1, 1),
354  (2, '2026-04-12', 'Stable', '2026-05-06', 'Migraine managed. Advised to avoid
355  triggeres and limit screen exposure', 2, 2),
356  (4, '2026-04-20', 'Stable', '2026-05-10', 'Febrile seizure resolved. EEG
357  follow-up scheduled in 2 weeks', 4, 4);
```